

Appendix: Choosing Better NLDL Layouts by Evaluating the Model in the High-dimensional Data Space

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1 Methods and hyper-parameters used to generate layouts

Table 1 contains the list of methods and hyper-parameters used for each of the layouts shown in the paper.

Figure	NLDR method	Hyper-parameter(s)
1a	UMAP	n_neighbors = 30, min_dist = 0.3
1b	UMAP	n_neighbors = 5, min_dist = 0.8
1c	UMAP	n_neighbors = 5, min_dist = 0.01
1d	tSNE	perplexity = 5
1e	tSNE	perplexity = 30
1f	PHATE	knn = 5
1g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
1h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
2	tSNE	perplexity = 47
4a	tSNE	perplexity = 47
5b	tSNE	perplexity = 47
6	tSNE	perplexity = 47
8a	tSNE	perplexity = 47
8b	tSNE	perplexity = 62
8c	UMAP	n_neighbors = 15, min_dist = 0.1
8d	PHATE	knn = 5
8e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3

(continued)

Figure	NLDR method	Hyper-parameter(s)
8f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
9a	tSNE	perplexity = 47
9b	tSNE	perplexity = 62
9c	UMAP	n_neighbors = 15, min_dist = 0.1
9d	PHATE	knn = 5
9e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
9f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
10a	UMAP	n_neighbors = 30, min_dist = 0.3
10b	UMAP	n_neighbors = 5, min_dist = 0.8
10c	UMAP	n_neighbors = 5, min_dist = 0.01
10d	tSNE	perplexity = 5
10e	tSNE	perplexity = 30
10f	PHATE	knn = 5
10g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
10h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
11a	UMAP	n_neighbors = 30, min_dist = 0.3
11e	tSNE	perplexity = 30
12a	tSNE	perplexity = 30
12b	tSNE	perplexity = 89
12c	UMAP	n_neighbors = 15, min_dist = 0.1
12d	PHATE	knn = 5

(continued)

Figure	NLDR method	Hyper-parameter(s)
12e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
12f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
13a	tSNE	perplexity = 30
14a	tSNE	perplexity = 30
A4a	tSNE	perplexity = 71
A4b	UMAP	n_neighbors = 15, min_dist = 0.1
A4c	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2
A5	tSNE	perplexity = 52
A6a	UMAP	n_neighbors = 30, min_dist = 0.3
A6b	tSNE	perplexity = 30
A7a	UMAP	n_neighbors = 30, min_dist = 0.3
A7b	tSNE	perplexity = 30
A8a	UMAP	n_neighbors = 30, min_dist = 0.3
A8b	tSNE	perplexity = 30
A9a	UMAP	n_neighbors = 30, min_dist = 0.3
A9b	UMAP	n_neighbors = 5, min_dist = 0.8
A9c	UMAP	n_neighbors = 5, min_dist = 0.01
A9d	tSNE	perplexity = 5
A9e	tSNE	perplexity = 30
A9f	PHATE	knn = 5

(continued)

Figure	NLDR method	Hyper-parameter(s)
A9g	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
A9h	PaCMAP	n_neighbors = 30, init = random, MN_ratio = 0.9, FP_ratio = 5
A10a	tSNE	perplexity = 30
A10b	tSNE	perplexity = 89
A10c	UMAP	n_neighbors = 15, min_dist = 0.1
A10d	PHATE	knn = 5
A10e	TriMAP	n_inliers = 12, n_outliers = 4, n_random = 3
A10f	PaCMAP	n_neighbors = 10, init = random, MN_ratio = 0.5, FP_ratio = 2

Table 1: NLDR methods and hyper-parameters used for each Figure in the main paper.

2 Videos links

Animations of the p - D tours that produced specific projections shown in some figures in the main paper are available on YouTube at the links given in Table 2.

Figure	URL
4	youtu.be/yHKTHK4UBiU
5	youtu.be/Fukiminr090
11	youtu.be/3VfK3M2gnZM , youtu.be/Es84bwQcndU
13	youtu.be/sUcGd57Swdg , youtu.be/QiklCjELUxo

Table 2: Videos of the langevitour animations and the linked plots.

3 Notation

Notation	Description
n, p, k	number of observations, variables, embedding dimension, respectively
$\mathbf{X}, \mathbf{x}$	p -dimensional data (population, sample)
$\mathbf{y}$	k -dimensional layout
P	orthonormal basis, generating a d -dimensional linear projection of p -dimensional data
T	true model
g	functional mapping from p -D to k -D, especially as prescribed by NLDR
θ	(Hyper-) parameters for NLDR method
r	ranges of the embedding components
$C^{(j)}$	j -dimensional bin centers
(b_1, b_2)	number of bins in each direction
(a_1, a_2)	binwidths, distance between centroids in each direction
(s_1, s_2)	starting coordinates of the hexagonal grid
q	buffer to ensure hexgrid covers data, proportion of data range, 0-1
m	number of non-empty bins
b	number of hexagons in the grid
h	hexagonal id
l	side length
A	area
n_h	bin count
w_h	standardised bin counts
d_h	bin density

Table 3: Summary of notation for describing new methodology.

4 Scripts

Folder	Script	Description
NA	additional_functions.R	NA
NA	evaluation.py	NA
NA	nldr_code.R	NA
two_nonlinear	01_gen_data.R	NA

(continued)

Folder	Script	Description
two_nonlinear	02_gen_true_model.R	NA
two_nonlinear	03_gen_embeddings.R	NA
two_nonlinear	04_gen_mse_for_diff_meth- ods.R	NA
two_nonlinear	05_gen_rm_lwd_mse.R	NA
two_nonlinear	06_gen_model_with_tSNE.R	NA
two_nonlinear	07_example_evaluation_metrics.R	NA
two_nonlinear	08_gen_model_with_PHATE.R	NA
five_gau_clusters	01_five_gaussian_cluster_data_emb.R	NA
five_gau_clusters	02_gen_model_with_tSNE.R	NA
five_gau_clusters	03_gen_model_with_UMAP.R	NA
five_gau_clusters	04_gen_model_with_PaCMAP.R	NA
one_c_shaped_dens_structure	01_gen_data.R	NA
one_c_shaped_dens_structure	02_gen_embeddings_uni_dens.R	NA
one_c_shaped_dens_structure	03_gen_model_with_tSNE.R	NA
pbmc3k	01_obtain_pca_author.R	NA

(continued)

Folder	Script	Description
pbmc3k	02_obtain_umap_authors.R	NA
pbmc3k	03_gen_umap_diff_param.R	NA
pbmc3k	04_gen_tsne_diff_param.R	NA
pbmc3k	05_gen_phate.R	NA
pbmc3k	06_gen_trimap.R	NA
pbmc3k	07_gen_pacmap.R	NA
pbmc3k	08_gen_mse_for_diff_methods.R	NA
pbmc3k	09_gen_scDEED.R	NA
pbmc3k	10_pre_process_for_embedding.R	NA
pbmc3k	11_gen_mse_for_diff_tsne_scDEED.R	NA
pbmc3k	12_gen_mse_for_diff_umap_scDEED.R	NA
pbmc3k	13_gen_model_with_UMAP.R	NA
pbmc3k	14_gen_model_with_tSNE.R	NA
pbmc3k	15_gen_model_with_UMAP_scDEED.R	NA
pbmc3k	16_gen_model_with_tSNE_scDEED.R	NA
pbmc3k	17_evaluation_metrics.R	NA
pbmc3k	18_evaluation_metrics_scDEED.R	NA
mnist	01_data_preprocessing.R	NA

(continued)

Folder	Script	Description
mnist	02_gen_diff_embeddings.R	NA
mnist	03_gen_mse_for_diff_meth- ods.R	NA
mnist	04_gen_model_with_tSNE.R	NA
mnist	05_evaluation_metrics.R	NA
mnist	06_link_brush_layout_e.R	NA

Table 4: R and Python script files used to generate outputs in the main paper.

5 Generating the 2NC7 data

This data is constructed by simulating two clusters, each consisting of 1000 observations. The C-shaped cluster is generated from $\theta \sim U(-3\pi/2, 0)$, $X_1 = \sin(\theta)$, $X_2 \sim U(0, 2)$ (adding thickness to the C), $X_3 = \text{sign}(\theta) \times (\cos(\theta) - 1)$, $X_4 = \cos(\theta)$. Observations lie on a 2- D manifold in 7- D . The other cluster is from $X_1 \sim U(0, 2)$, $X_2 \sim U(0, 3)$, $X_3 = -(X_1^3 + X_2)$, and $X_4 \sim U(0, 2)$. It is also curved, but observations lie on a 3- D manifold in 7- D . Three more variables, X_5, X_6, X_7 , that are small amounts of pure noise are added. We would consider $T = (X_1, X_2, X_3, X_4)$ to be the geometric structure (true model) that we hope to capture.

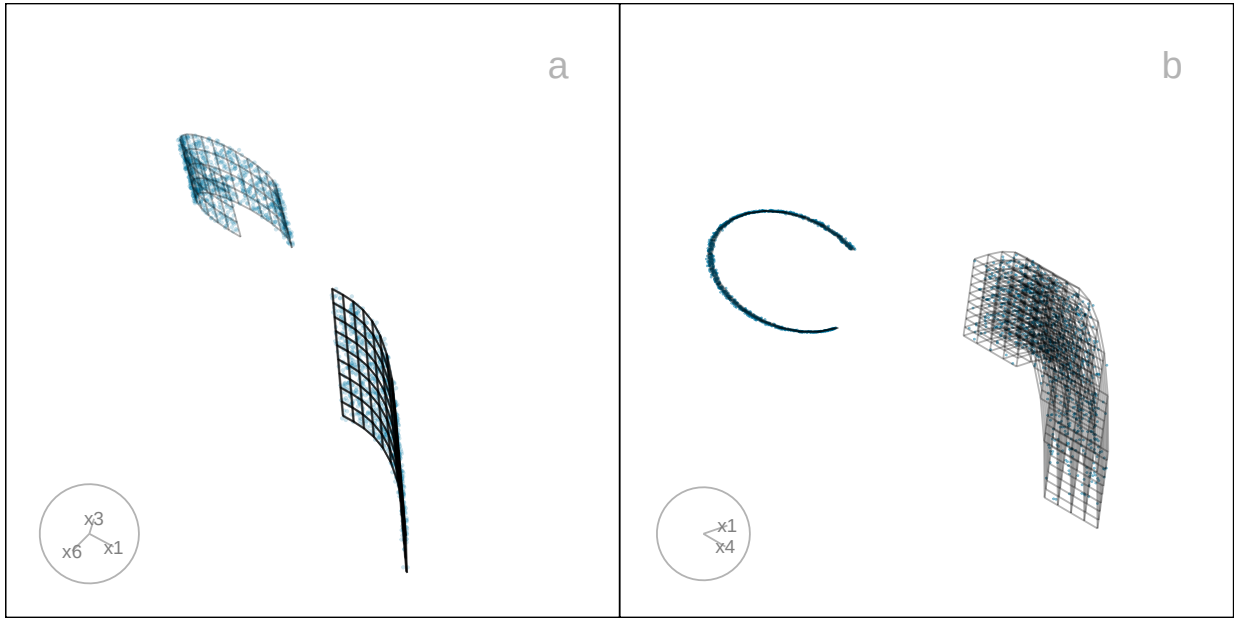


Figure 1: Two projections of the p -D true model overlaying the data are shown in a, b. Video of the langevitour animations is available at <https://youtu.be/35TrnYJsUUI>.

6 Computing hexagon grid configurations

Given range of embedding component, r_2 , number of bins along the x-axis, b_1 , and buffer proportion, q , hexagonal starting point coordinates, $s_1 = -q$, and $s_2 = -qr_2$. The purpose is to find width of the hexagon. a_1 , and number of bins along the y-axis, b_2 .

Geometric arguments give rise to the following constraints.

$\min a_1$ s.t.

$$s_1 - \frac{a_1}{2} < 0, \quad (1)$$

$$s_1 + (b_1 - 1) \times a_1 \geq 1, \quad (2)$$

$$s_2 - \frac{a_2}{2} < 0, \quad (3)$$

$$s_2 + (b_2 - 1) \times a_2 \geq r_2. \quad (4)$$

Since a_1 and a_2 are distances,

$$a_1, a_2 > 0.$$

Also, $(s_1, s_2) \in (-0.1, -0.05)$ as these are multiplicative offsets in the negative direction.

Equation 1 can be rearranged as,

$$a_1 > 2s_1$$

which given $s_1 < 0$ and $a_1 > 0$ will *always* be true. The same logic follows for Equation 3

and substituting $a_2 = \sqrt{3}a_1/2$, and $s_2 = -qr_2$ to Equation 3 can be written as,

$$a_1 > -\frac{4}{\sqrt{3}}qr_2$$

Also, substituting $a_2 = \sqrt{3}a_1/2$, $s_2 = -qr_2$ and rearranging Equation 4 gives:

$$a_1 \geq \frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)}. \quad (5)$$

Similarly, substituting $s_1 = -q$ Equation 2 becomes,

$$a_1 \geq \frac{(1+q)}{(b_1-1)}. \quad (6)$$

This is a linear optimization problem. Therefore, the optimal solution must occur on a vertex. So, by setting Equation 5 equals to Equation 6 gives,

$$\frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)} = \frac{(1+q)}{(b_1 - 1)}.$$

After rearranging this,

$$b_2 = 1 + \frac{2r_2(b_1 - 1)}{\sqrt{3}}$$

and since b_2 should be an integer,

$$b_2 = \left\lceil 1 + \frac{2r_2(b_1 - 1)}{\sqrt{3}} \right\rceil. \quad (7)$$

Furthermore, with known b_1 and b_2 , by considering Equation 2 or Equation 4 as the *binding* or *active constraint*, can compute a_1 .

If Equation 2 is active, then,

$$\frac{(1+q)}{(b_1 - 1)} < \frac{2(r_2 + qr_2)}{\sqrt{3}(b_2 - 1)}.$$

Rearranging this gives,

$$r_2 > \frac{\sqrt{3}(b_2 - 1)}{2(b_1 - 1)}.$$

Therefore, if this equality is true, then

$$a_1 = \frac{(1+q)}{(b_1-1)},$$

otherwise,

$$a_1 = \frac{2r_2(1+q)}{\sqrt{3}(b_2-1)}.$$

7 Binning the data

Points are assigned to the bin they fall into based on the nearest centroid. If a point is equidistant from multiple centroids, it is assigned to the centroid with the smallest bin ID.

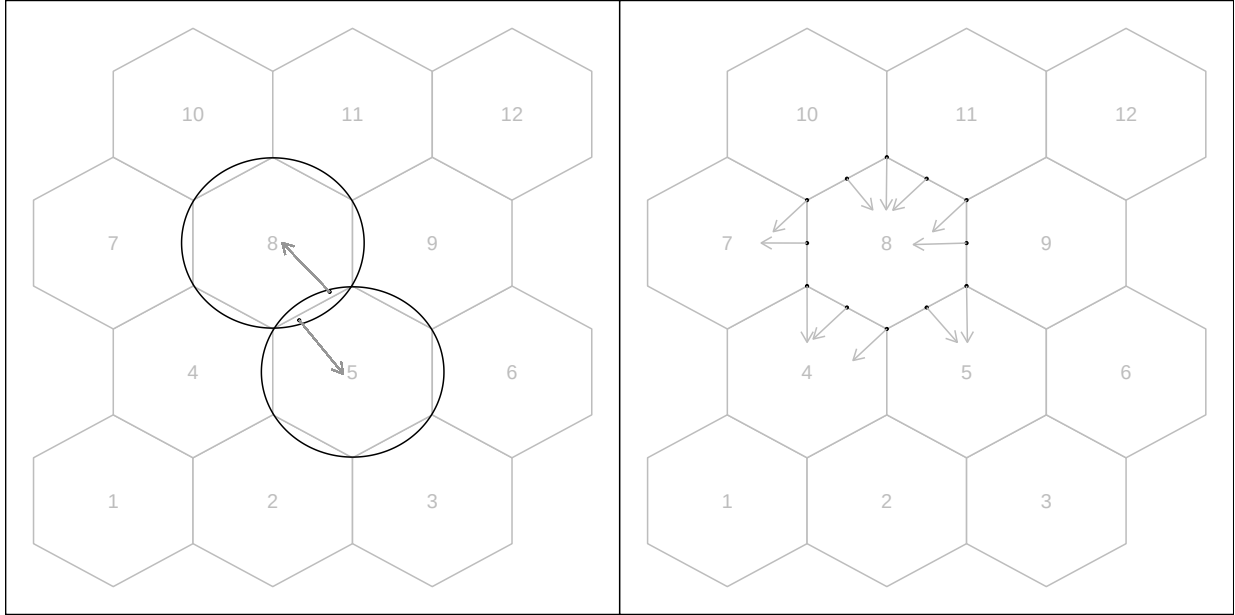


Figure 2: Binning the data. Points are assigned to the nearest centroid. If a point is equidistant from multiple centroids, assigned to the centroid with the smallest bin ID.

8 Area of a hexagon

The area of a hexagon is defined as $A = 3\sqrt{3}l^2/2$, where l is the side length of the hexagon.

l can be computed using a_1 and a_2 .

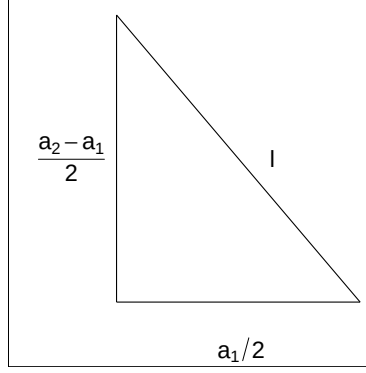


Figure 3: The components of the right triangle illustrating notation.

By applying the Pythagorean theorem, we obtain,

$$l^2 = \left(\frac{a_1}{2}\right)^2 + \left(\frac{a_2 - l}{2}\right)^2.$$

Next, rearranging the terms, we get,

$$l^2 - \left(\frac{a_2 - l}{2}\right)^2 = \left(\frac{a_1}{2}\right)^2,$$

$$\left[l - \left(\frac{a_2 - l}{2}\right)\right] \left[l + \left(\frac{a_2 - l}{2}\right)\right] = \left(\frac{a_1}{2}\right)^2,$$

$$3l^2 + 2a_2l - (a_1^2 + a_2^2) = 0.$$

Finally, by solving the quadratic equation, we compute,

$$l = \frac{-2a_2 \pm \sqrt{4a_2^2 - 24[-(a_1^2 + a_2^2)]}}{6},$$

$$l = \frac{-a_2 \pm \sqrt{a_2^2 - 6[-(a_1^2 + a_2^2)]}}{3},$$

where $l > 0$.

9 Curiosities about NLDR results discovered by examining the model in the data space

With the drawing of the model in the data, several interesting differences between NLDR methods can be observed.

9.1 Some methods appear to order points in the layout

The 2- D model representations generated from some NLDR methods, especially PaCMAP, are unreasonably flat or like a pancake. A simple example of this can be seen with data simulated to contain five 4- D Gaussian clusters. Each cluster is essentially a ball in 4- D , so there is no 2- D representation, rather the model in each cluster should resemble a crumpled sheet of paper that fills out 4- D .

Figure 4 a1, b1, c1 show the 2- D layouts for (a) tSNE, (b) UMAP, and (c) PaCMAP, respectively. The default hyper-parameters for each method are used. In each layout we can see an accurate representation where all five clusters are visible, although with varying degrees of separation.

The models are fitted to each these layouts. Figure 4 a2, b2, c2 show the fitted models in a projection of the 4- D space, taken from a tour. These clusters are fully 4- D in nature, so we would expect the model to be a *crumpled sheet* that stretches in all four dimensions. This is what is mostly observed for tSNE and UMAP. The curious detail is that the model for PaCMAP is closer to a *pancake* in shape in every cluster! This single projection only shows this in three of the five clusters but if we examine a different projection the other clusters exhibit the pancake also. While we don't know what exactly causes this, it is likely due to some ordering of points in the 2- D PaCMAP layout that induces the flat model. One could imagine that if the method used principal components on all the data, that it might induce some ordering that would produce the flat model. If this were the reason, the pancaking would be the same in all clusters, but it is not: The pancake is visible in some clusters in some projections but in other clusters it is visible in different projections. It might be due to some ordering by nearest neighbors in a cluster. The PaCMAP documentation doesn't provide any helpful clues. That this happens, though, makes the PaCMAP layout inadequate for representing the high-dimensional data.

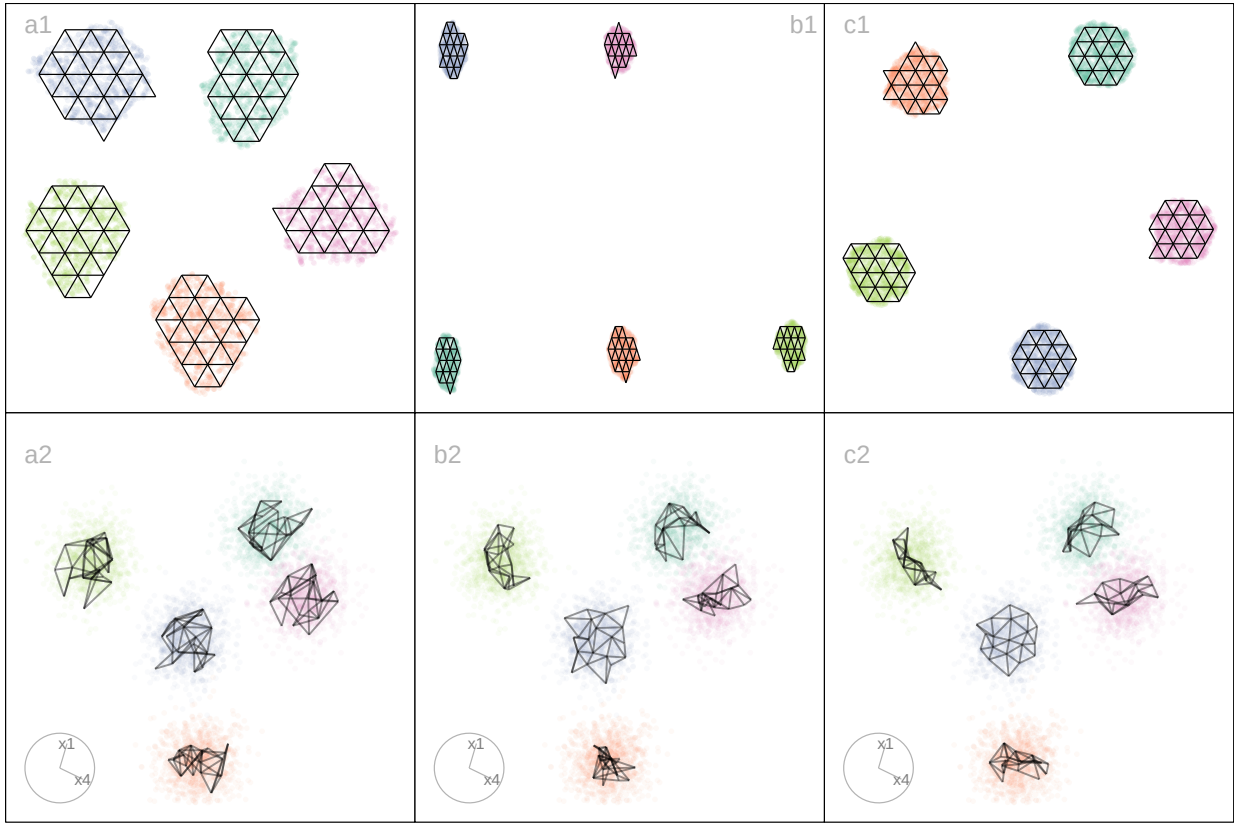


Figure 4: NLDR’s organise points in the 2- D layout in different ways, possibly misleadingly, illustrated using three layouts: (a) tSNE, (b) UMAP, (c) PaCMAP. The data has five Gaussian clusters in 4- D . The bottom row of plots shows a 2- D projection from a tour on 4- D revealing the differences generated by the layouts on the model fits. We would expect the model fit to be like that in (a2) where it is distinctly is separate for each cluster but like a hairball in each. This would indicate the distinct clusters, each being fully 4- D . With (c2), the curiosity is that the model is a 2- D pancake shape in 4- D , indicating that there is some ordering of points done by PaCMAP, possibly along some principal component axes. Videos of the langevitour animations are available at <https://youtu.be/I-kxCwVfqiQ>, <https://youtu.be/gD1P01FUPyU>, and https://youtu.be/MxJ_srOFQNk respectively.

9.2 Sparseness creates a contracted 2- D layout

Differences in density can arise by sampling at different rates in different subspaces of p - D . For example, the data shown in Figure 5 all lies on a 2- D curved sheet in 4- D , but one end of the sheet is sampled densely and the other very sparsely. It was simulated to illustrate the effect of the density difference on layout generated by an NLDR, illustrated using the tSNE results, but it happens with all methods.

Figure 5 (a2, b2) shows a 2- D layout for tSNE created using the default hyper-parameters. One would expect to see a rectangular shape if the curved sheet is flattened, but the layout is triangular. The other two displays show the residuals as a dot density plot (a1, b1), and a 2- D projection of the data and the model from 4- D (a3, b3). Using linked brushing between the plots, we can highlight points in the tSNE layout, and examine where they fall in the original 4- D . The darker (maroon) points indicate points that have been highlighted by linking. In row a, the points at the top of the triangle are highlighted, and we can see these correspond to higher residuals, and also to all points at the low density end of the curved sheet. In row b, points at the lower left side of the triangle are highlighted which corresponds to smaller residuals and one corner of the sheet at the high density end of the curved sheet.

The tSNE behaviour is to squeeze the low density area of the data together into the layout. This is common in other NLDR methods also, which means analysts need to be aware that if their data is not sampled relatively uniformly, apparent closeness in the 2- D may correspond to sparseness in p - D .

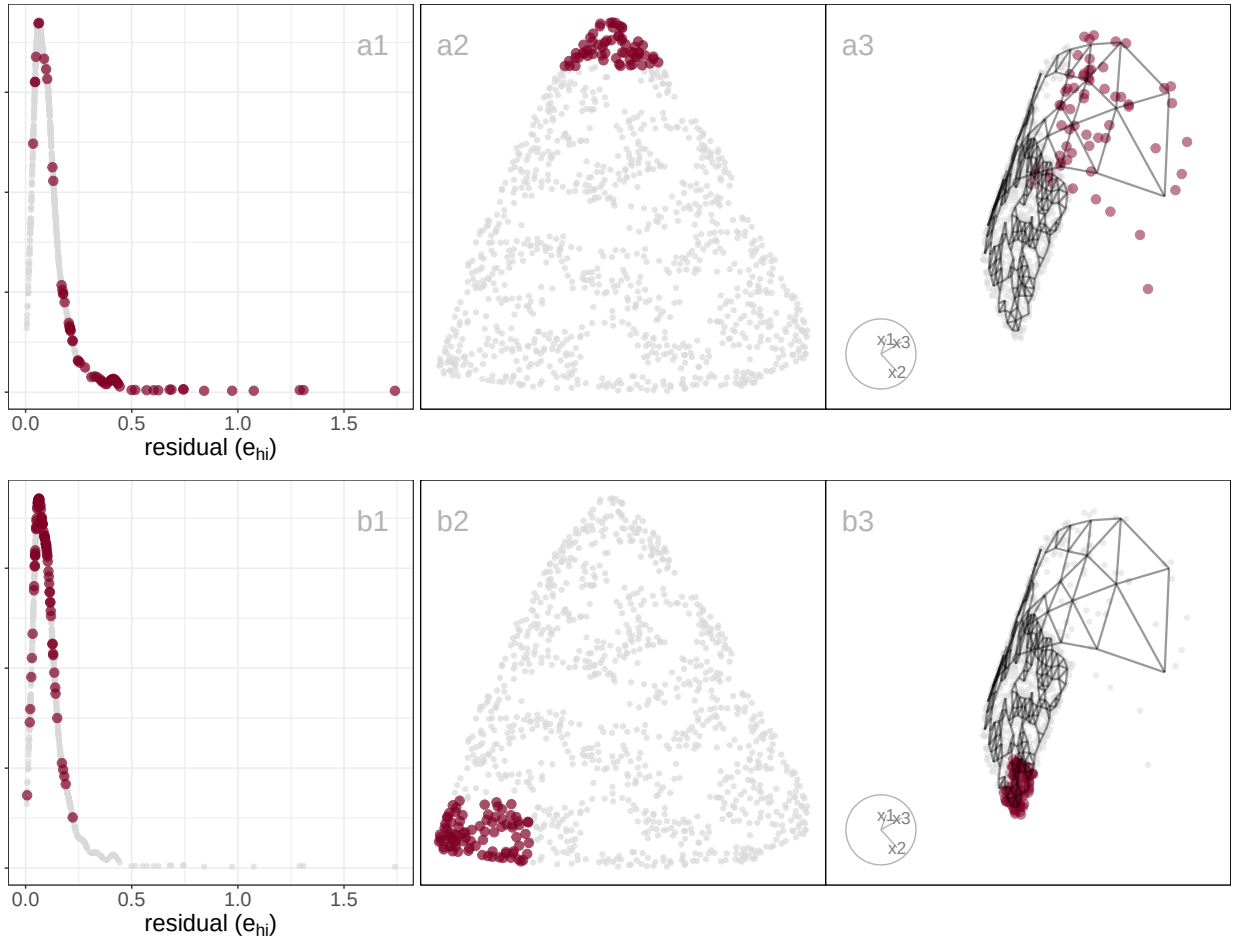


Figure 5: Exploring the effect of density on the NLDR layout using a 2- D curved sheet in 4- D with different density at each end. Three plots are linked: density plot of residuals (a1, b1), NLDR layout (a2, b2), projection of 4- D model and data (a3, b3). The brown points indicate the selected set, which are different in each row. In (a2), the top part of the triangular shape is selected which corresponds to higher residuals (a1) and the sparse end of the structure (a3). In (b2) one of other corners is highlighted, which can be seen to correspond to low residuals (b1) and one side of the dense end of the data (b3). While the tSNE layout represents the dense end of the sheet correctly as two corners in the layout, it contracts the sparse end of the sheet into a single corner. Video of the langevitour animation is available at <https://youtu.be/-KsQH0rII2A>.

10 Single-cell gene expression: comparison with results of scDEED recommendations

As we were writing this paper [Xia et al. \(2023\)](#) appeared proposing a new method called scDEED helping to assess the validity of a 2- D embedding. scDEED calculates a reliability score for each cell embedding based on the similarity between the cell’s 2- D embedding neighbors and its neighbors prior to embedding. A low reliability score suggests a dubious embedding. It can help in the deciding on optimal hyper-parameters. Here we illustrate how our method compares with the results from scDEED.

Note that [Xia et al. \(2023\)](#) uses a different PBMC dataset than that used by [Chen et al. \(2024\)](#), shown by us in the main paper example, which is why this comparison is shown here and not in the main paper. Their data contains 31,021 cells including cell type labels, and the gene expression levels were in the unit of log-transformed UMI count per 10,000. They focused on three sequencing methods (inDrops, DropSeq, and SeqWell) and four common cell types Cytotoxic T cell, CD4+T cell, CD14+ Monocyte, and B cell. Pre-processing follows the process in [Xia et al. \(2023\)](#) again using the Human Peripheral Blood Mononuclear Cells (PBMC) data.

For illustration purposes, we only selected cells generated with inDrops ($n = 5858$ cells). Also, [Xia et al. \(2023\)](#) used first 9 principal components to generate the UMAP and tSNE with default hyper-parameters. The objective is to assess the optimized layout by scDEED, and if it does not accurately represent the three clusters with small separations of the PBMC dataset, then select a reasonable 2- D layout.

The layout a (Figure 6) is generated from the hyper-parameters suggested by [Chen et al. \(2024\)](#), and the layout b (Figure 6) is with suggested hyper-parameters by scDEED to be

more accurate. The layout a and b contain 46 and 83 dubious cells respectively.

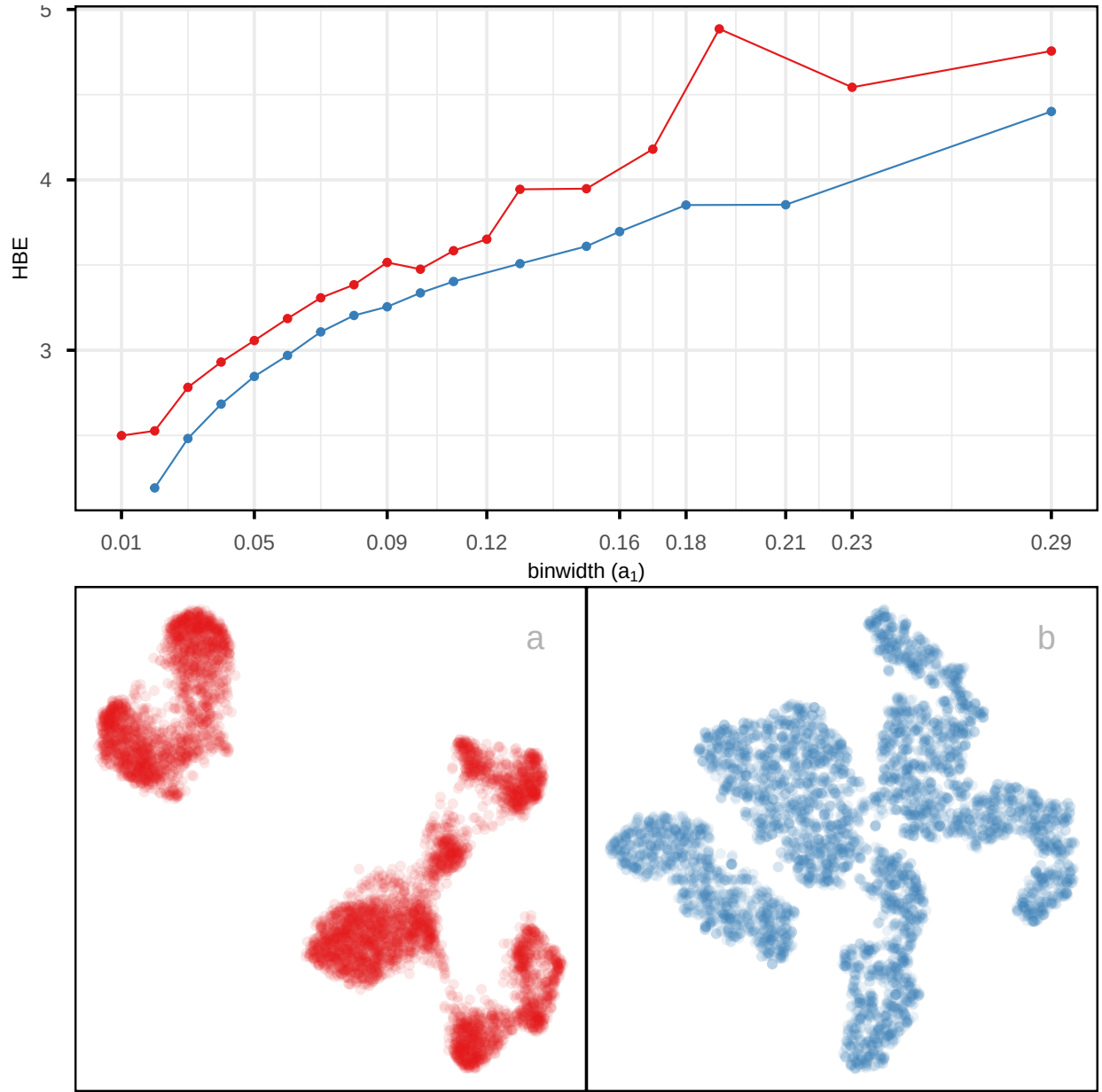
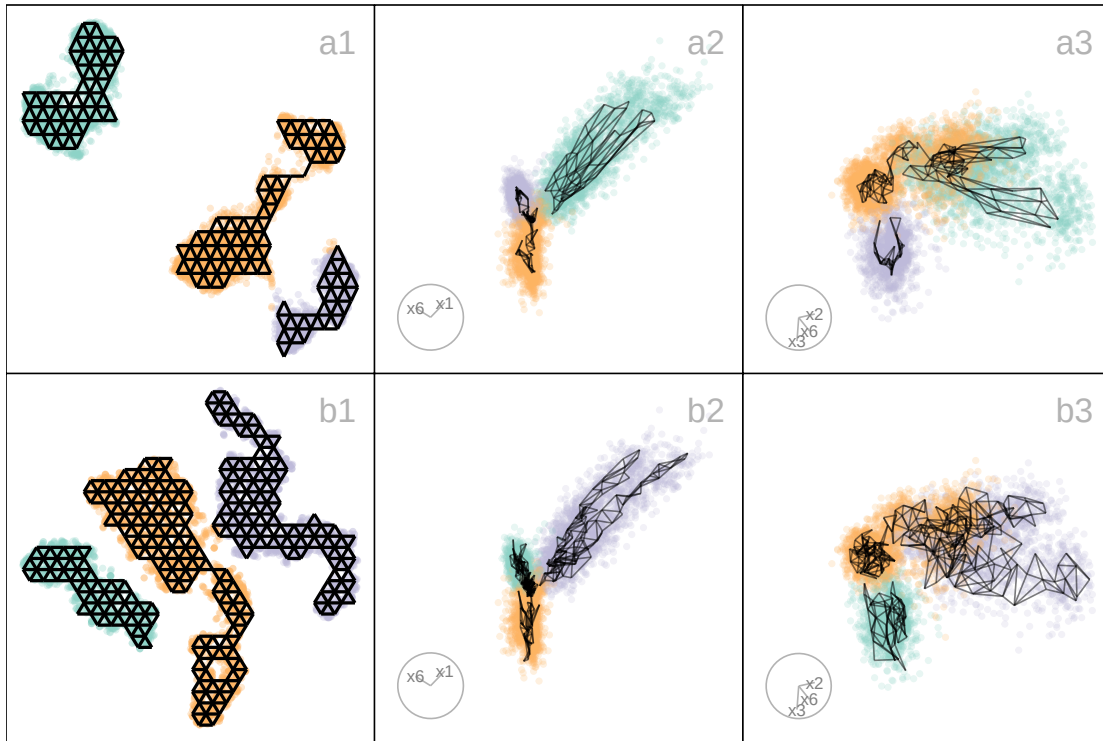


Figure 6: Assessing which of the 2 NLDR layouts on the PBMC3k data is the better representation using HBE for varying binwidth (a_1). Color used for the lines and points in the left plot and in the scatterplots represents NLDR layouts (a, b). Of the two, layout b is optimal across all binwidths making it the best choice.

The HBE vs binwidth (a_1) plot (Figure 6) illustrates that our approach would suggest that scDEED is correct here, that layout b is more accurately reflecting the cluster structure in the PBMC data.



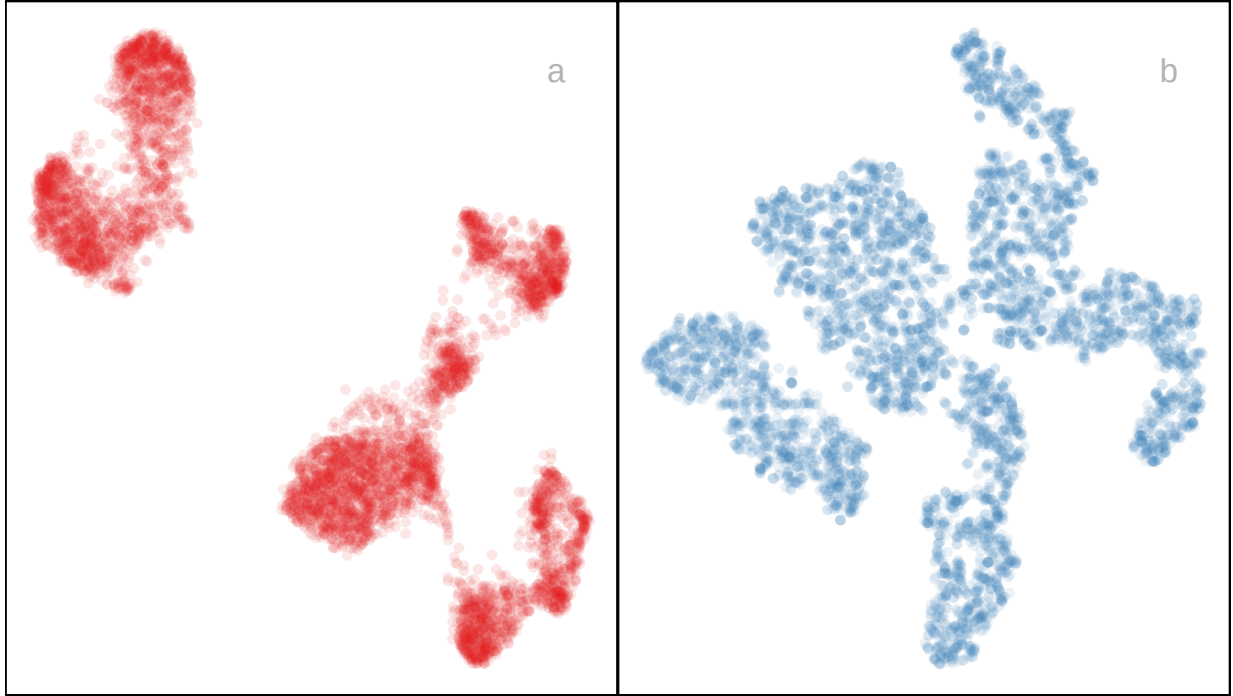
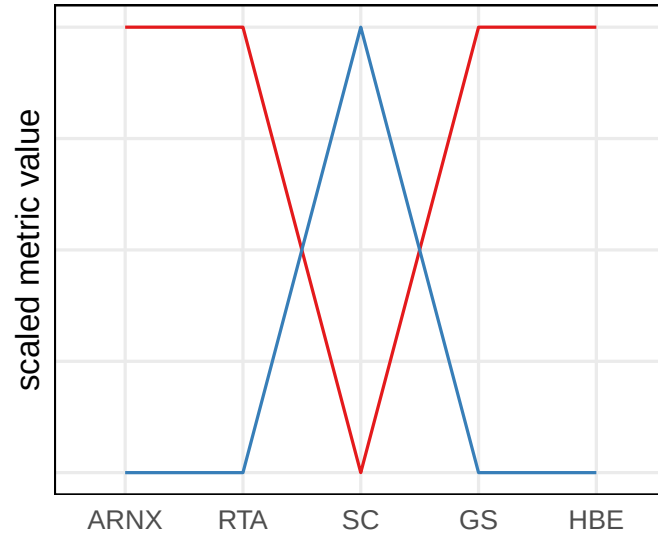


Figure 8: Comparison of evaluation metrics across eight NLDR layouts for the MNIST digit 1 data. Each line represents a layout evaluated using RTA, ARNX, sc, GS, and HBE. The metrics show different patterns, meaning no single measure tells the full story of embedding quality. Interestingly, the brown layout, rated the best by ARNX, sc, and GS, is seen as the worst by HBE, while the green layout, rated best by RTA, is seen as the best by HBE as well.

11 Compare HBE with existing evaluation metrics

Figure 9 and Figure 10 compare HBE with commonly used evaluation metrics such as RTA, ARNX, sc, and GS across multiple NLDR layouts. These visual comparisons highlight that HBE behaves differently from these existing metrics due to the different settings involved.

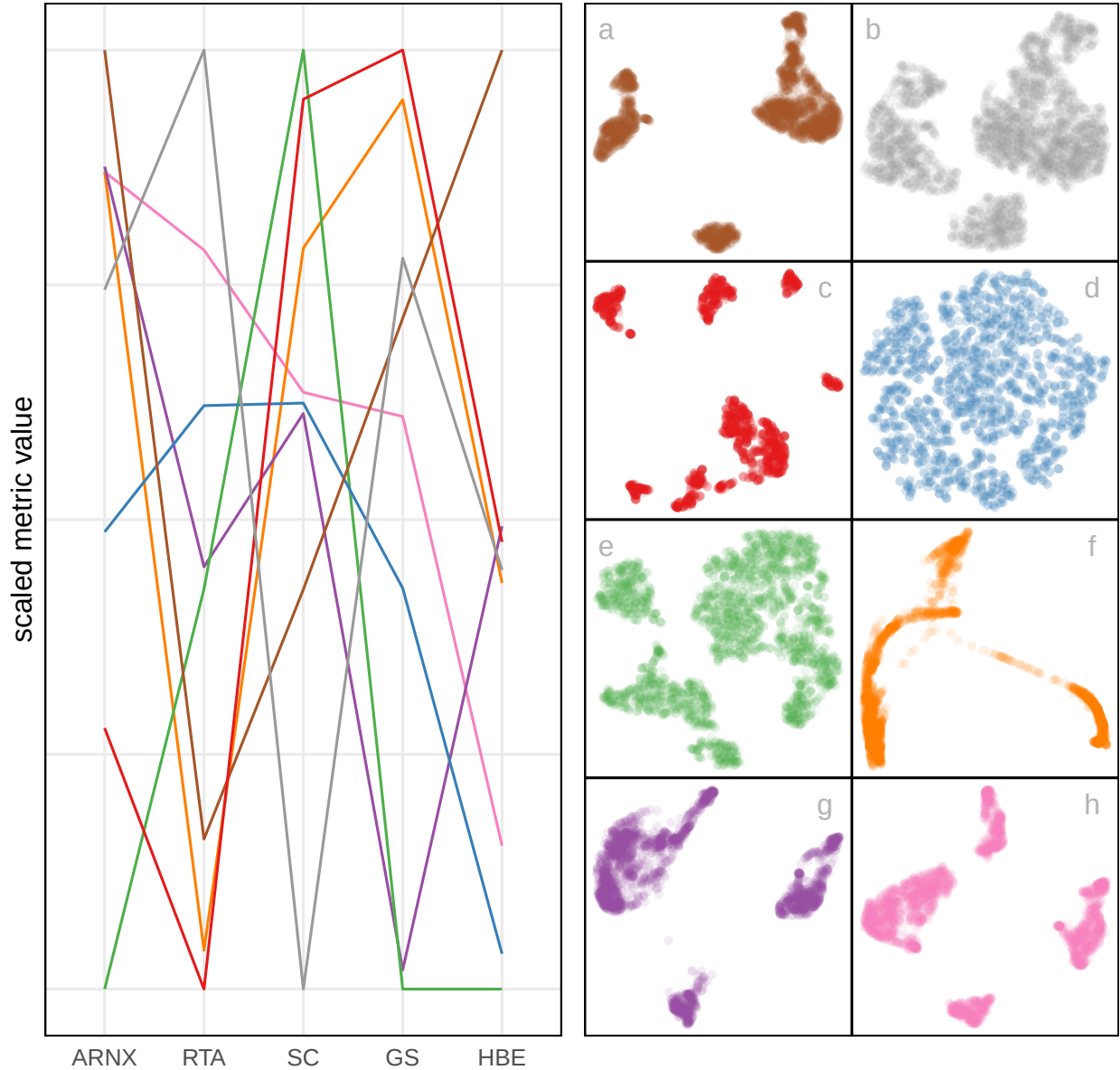


Figure 9: Comparison of evaluation metrics across eight NLDR layouts for the PBMC3k data. Each line represents a layout evaluated using RTA, ARNX, sc, GS, and HBE. The metrics show different patterns, meaning no single measure tells the full story of embedding quality. Interestingly, the green layout, rated worst by ARNX and GS, is seen as the best by HBE and sc, while the brown layout, rated best by ARNX, is seen as the worst by HBE showing that HBE captures a different view of how well the data structure is represented.

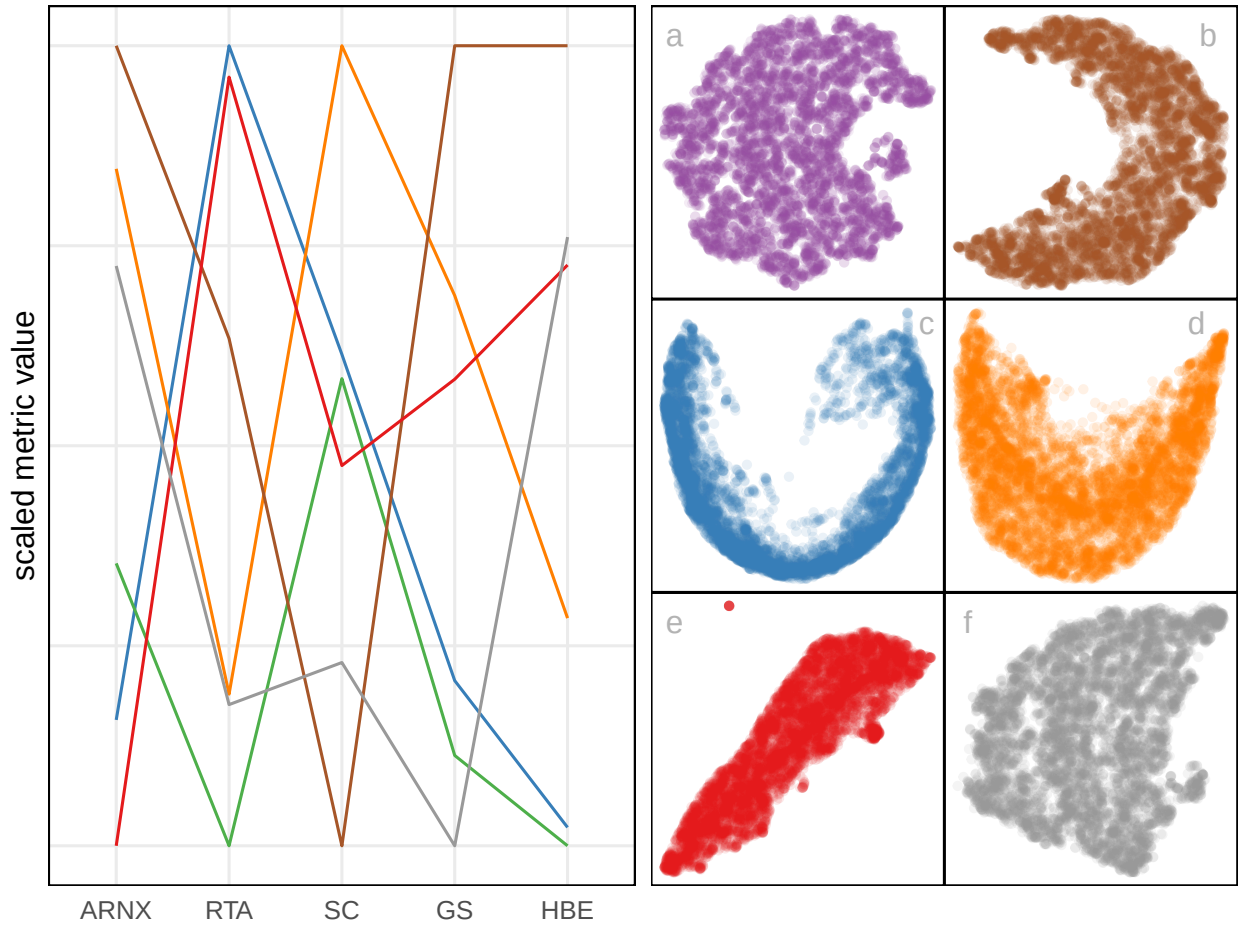


Figure 10: Comparison of evaluation metrics across eight NLDR layouts for the MNIST digit 1 data. Each line represents a layout evaluated using RTA, ARNX, sc, GS, and HBE. The metrics show different patterns, meaning no single measure tells the full story of embedding quality. Interestingly, the brown layout, rated the best by ARNX, sc, and GS, is seen as the worst by HBE, while the green layout, rated best by RTA, is seen as the best by HBE as well.

References

- Chen, Z., Wang, C., Huang, S., Shi, Y. & Xi, R. (2024), ‘Directly Selecting Cell-type Marker Genes for Single-cell Clustering Analyses’, *Cell Reports Methods* 4(7), 100810. <https://www.sciencedirect.com/science/article/pii/S2667237524001735>.
- Xia, L., Lee, C. & Li, J. J. (2023), ‘scDEED: A Statistical Method for Detecting Dubious

2D Single-cell Embeddings', *bioRxiv* . <https://www.biorxiv.org/content/early/2023/04/25/2023.04.21.537839>.